Udaychandra Gollapally

(303) 961-0160 | udaychandragollapally11@gmail.com | LinkedIn | GitHub

SUMMARY

Entry-level Software Engineer proficient in Python, Django, and Flask with demonstrated expertise in writing maintainable, production-ready code. Developed scalable microservices and achieved 90%+ test coverage while optimizing performance using design patterns and cloud infrastructure. Skilled in leveraging data science libraries and backend frameworks to build business-critical systems. Eager to contribute technical proficiency and a continuous learning mindset to drive data-driven decisions.

EDUCATION

California State University, Chico — **Aug 2023 - May 2025**
MS, Computer Science

CMR College of Engineering & Technology,Hyderabad — **Jul 2018 - May 2022**
B.Tech., Computer Science & Engineering

PROFESSIONAL EXPERIENCE

Planon Software Services Pvt. Ltd. — **Aug 2022 - Jul 2023**
Software Engineer - Backend — *Hyderabad, India*

- Developed scalable Java Spring Boot microservices for enterprise licensing, optimizing throughput by 25% on datasets >10M records while leveraging distributed systems principles.
- Built secure, performant RESTful APIs using Spring Boot, consumed by Wicket-based UIs, which reduced response times by ~30ms.
- Implemented Hibernate ORM for persistence; wrote optimized SQL for batch operations, improving query performance by 40%.
- Integrated OAuth 2.0, JWT authentication, and centralized configuration using Spring Cloud Config with Resilience4j, securing microservices and enhancing fault tolerance across environments.
- Automated CI/CD pipelines with Jenkins and Docker; enabled zero-downtime deployments via staged rollouts and rigorous health checks.
- Maintained 90%+ test coverage using JUnit/Mockito by writing maintainable, production-ready code and enforcing SOLID principles through peer reviews and static code analysis.

Planon Software Services Pvt. Ltd. — **Sep 2021 - Jul 2022**
Software Engineer Intern — *Hyderabad, India*

- Designed and built full-stack modules with Spring Boot (REST APIs), Hibernate (ORM), and Apache Wicket (UI), applying dependency injection and clean architecture for modular and maintainable design.
- Optimized native SQL and HQL queries for reporting systems, reducing execution time by 35%.
- Collaborated cross-functionally to transform UI wireframes into dynamic JSP/Servlet-based components, ensuring seamless integration and improved user experience.

Synergina Pvt. Ltd — **May 2021 - Sep 2021**
Data Science Intern — *Hyderabad, India*

- Developed predictive models with Python and scikit-learn, improving accuracy by 20% through targeted feature engineering and rigorous testing to achieve production quality.
- Automated ETL workflows and web scraping pipelines with BeautifulSoup and Scrapy, reducing manual effort by 80% and streamlining data processing.
- Constructed stakeholder-facing dashboards using Matplotlib and Seaborn to visualize key metrics, facilitating data-driven decision-making.

PROJECTS

AI-Driven Prediction of Early Readmission in Diabetic Patients | XGBoost, Random Forest, SHAP, LIME:

- Co-authored a research paper proposing a stacking ensemble model combined with explainable AI (XAI) techniques to predict early-hospital readmissions for diabetic patients
- Integrated SMOTE for class imbalance, deployed interpretable predictions using SHAP and LIME
- Achieved 92.5% accuracy and 0.975 ROC-AUC on real-world clinical datasets
- Published as part of a peer-reviewed conference submission

Django Chess | Javascript, Python, CSS, HTML:

- Implemented a multiplayer Django chess game with features like Invitations, Game History. Uses Web Sockets for real-time Synchronization of game state.

Yolo Traffic Sign Detection | Python, ultralytics, YOLO:

- A simple YOLO-based object detection project to identify traffic signs.

Car Price Prediction API| Python, scikit-learn, Flask:

- Created an ML pipeline and deployed a REST API in Docker for real-time resale value predictions.

Certification Verification System| Java, Spring Boot, MySQL:

- Developed a microservice to authenticate and verify professional certifications with JWT-based security and audit logging.

Efficient Retrieval-Augmented Generation (RAG) System | Python, PyTorch, NLP, Machine Learning:

- Built end-to-end local RAG pipeline using small language models (MiniLM, DistilGPT2) achieving 87% accuracy with 100% offline deployment for privacy-sensitive applications
- Engineered custom sliding window chunking with 30% overlap, reducing hallucination rates from "Very High" to "Low" while preserving semantic context across 1,200-page documents
- Implemented efficient PyTorch vector search using cosine similarity to handle 100k+ documents with <500MB RAM and sub-second inference times, replacing expensive FAISS infrastructure
- Achieved 47% accuracy improvement over baseline GPT-2 through dense vector retrieval, demonstrating small models with context outperform large models without retrieval

Handwritten Digit Recognition with Custom Generalization Testing : Python, PyTorch, MNIST, Data Augmentation, MLP, Confusion Matrix

- Developed a fully connected neural network (MLP) for handwritten digit classification using PyTorch; achieved 99.15% accuracy on MNIST.
- Designed custom test sets with visual distortions and augmentations to test model generalization under real-world variation.
- Applied data augmentation techniques like affine transforms, rotation, jittering to boost robustness.
- Trained four models using fine-tuning strategies, resulting in a 3× accuracy improvement on distorted data (from 17% to 53%).
- Evaluated model performance using precision, recall, and confusion matrices; highlighted weak-performing digit classes to guide improvements.

TECHNICAL SKILLS

- **Programming Languages and Frameworks**: C++, Python, Java, SQL, JavaScript, Django, Spring Boot, Hibernate, Apache Wicket
- **Cloud Platforms and DevOps**: Google Cloud Platform (GCP), CI/CD, Docker, Git, Maven
- **Database Management**: Relational Databases, Query Optimization
- **Development and Integration**: RESTful APIs, SOAP Web Services, JSON, XML, Data Structures, Distributed Systems
- **Data and Machine Learning**: Pandas, NumPy, Matplotlib, Beautiful Soup, Scrapy, scikit-learn, Machine Learning
- **Version Control System**: Git
- **Software Testing and Debugging**: JUnit
- **Professional Skills**: Technical Decision Impact
- **Software Development Principles & Methodologies:**: Object-Oriented Programming (OOP), Software Development Life Cycle, Design Patterns, Agile Development